

LECTURE 8

Question:

What is the difference between the following?

$$\frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \quad \left| \begin{array}{l} |+\rangle \text{ with probability } \frac{1}{2} \\ |-\rangle \text{ " " " " } \end{array} \right.$$

$$N = 2^n$$

We want to interpret $\{0,1\}^n$ as $\mathbb{Z}_N$ and

look at $g: \mathbb{Z}_N \rightarrow \mathbb{C}$

$$|g\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} g(0) \\ g(1) \\ \vdots \\ g(N-1) \end{bmatrix} = \frac{1}{\sqrt{N}} \left(g(0) |0\rangle + g(1) |1\rangle + \dots + g(N-1) |N-1\rangle \right)$$

We assume $|g(0)|^2 + \dots + |g(N-1)|^2 = N$

The fourier transform allows expressing any function as a combination of linear function.

$$g(x) = \frac{1}{\sqrt{N}} (g(0)|0\rangle + \dots + g(N-1)|N-1\rangle)$$

$$= \hat{g}(0)|\chi_0\rangle + \hat{g}(1)|\chi_1\rangle + \dots + \hat{g}(N-1)|\chi_{N-1}\rangle$$

$$|\chi_i\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ \omega_N^i \\ \omega_N^{2i} \\ \vdots \\ \omega_N^{i(N-1)} \end{bmatrix}$$

$$\underline{\chi_i(x) = \omega_N^{ix}}$$

$$|\chi_i(x)| = 1$$

$$\begin{aligned}
\langle \chi_s | \chi_t \rangle &= \frac{1}{N} \sum_{x=0}^{N-1} \chi_s^*(x) \chi_t(x) \\
&= \frac{1}{N} \sum_{x=0}^{N-1} \omega_N^{-sx} \cdot \omega_N^{tx} \\
&= \frac{1}{N} \sum_{x=0}^{N-1} \omega_N^{(t-s)x}
\end{aligned}$$

$$t-s = a \neq 0$$

$$S = \omega_N^0 + \omega_N^a + \omega_N^{2a} + \dots + \omega_N^{(N-1)a}$$

$$d = \text{g.c.d.}(a, N)$$

$$\left(\frac{N}{d}\right)^{\text{th}} \text{ roots of unity are: } \underbrace{1, \omega_N^d, \omega_N^{2d}, \dots, \omega_N^{(\frac{N}{d}-1)d}}$$

$$a = k \cdot d$$

$$S = 1 + \omega_N^{kd} + \omega_N^{2kd} + \dots + \omega_N^{(\frac{N}{d}-1)kd} + \omega_N^{\frac{N}{d} \cdot kd} + \dots$$

$$t-s \neq 0 \Rightarrow \sum_{x=0}^{N-1} \omega_N^{(t-s)x} = 0$$

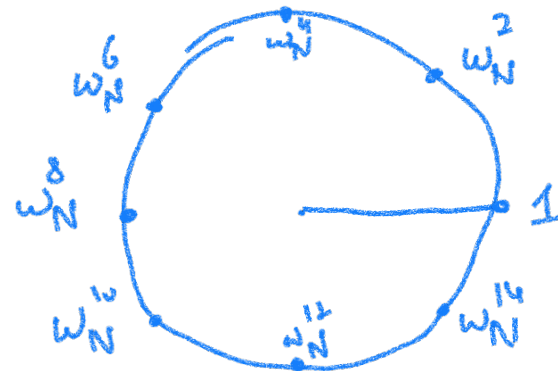
$$t-s=0 \Rightarrow \sum_{x=0}^{N-1} \omega_N^{0 \cdot x} = \sum_{x=0}^{N-1} 1 = N$$

$$\langle \chi_s | \chi_t \rangle = \begin{cases} 0 & \text{if } s \neq t \\ 1 & \text{if } s = t \end{cases}$$

$$N = 16$$

$$a = 6$$

$$1 + \omega_N^6 + \omega_N^{12} + \omega_N^2 + \omega_N^8 + \omega_N^{14} + \omega_N^4 + \omega_N^{10}$$



So $|x_0\rangle, \dots, |x_{N-1}\rangle$ are orthonormal

$$|s\rangle \xrightarrow{U} |x_s\rangle$$

$$U = \begin{bmatrix} |x_0\rangle & |x_1\rangle & \dots & |x_{N-1}\rangle \end{bmatrix}$$

$$g: \mathbb{Z}_N \rightarrow \mathbb{C} \quad \sum |g(i)|^2 = N$$

$$|g\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} g(0) \\ g(1) \\ \vdots \\ g(N-1) \end{bmatrix} = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} g(i) |i\rangle$$

$$|g\rangle \xrightarrow{\text{DFT}_N} \sum_s \hat{g}(s) |s\rangle$$

" $\sum_s \hat{g}(s) \underline{|x_s\rangle}$

$$\begin{aligned} \hat{g}(s) &= \langle x_s | g \rangle \\ &= \frac{1}{N} \sum_z x_s^*(z) g(z) \\ &= \frac{\sum_z \omega_N^{-sz} g(z)}{N} \end{aligned}$$

$$\sum_s \hat{g}(s) |s\rangle \longrightarrow \boxed{\text{DFT}^{-1}} \longrightarrow \sum_s \hat{g}(s) |x_s\rangle$$

$$\text{DFT}^{-1} = \begin{bmatrix} |x_0\rangle & |x_1\rangle & \dots & |x_{N-1}\rangle \end{bmatrix}$$

$$\sum_s \hat{g}(s) |s\rangle \longrightarrow \boxed{\text{DFT}^{-1}} \longrightarrow \sum_s \hat{g}(s) |\chi_s\rangle$$

$$\text{DFT}^{-1} \quad |s\rangle \longrightarrow |\chi_s\rangle$$

$$\text{DFT} \quad |s\rangle \longrightarrow |\chi_s^*\rangle$$

Recall $U^\dagger U = \text{Id.}$
 $U^\dagger = U^{-1}$

$$\text{DFT} = \begin{bmatrix} x_0^* & x_1^* & \dots & x_{N-1}^* \end{bmatrix}$$

$$DFT = \begin{bmatrix} 1 & \omega_N^0 & \omega_N^0 & \dots & \omega_N^0 \\ 1 & \omega_N^1 & \omega_N^2 & \dots & \omega_N^{N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_N^{N-1} & \omega_N^{2(N-1)} & \dots & \omega_N^{(N-1)^2} \end{bmatrix}$$

How to implement DFT.

Consider $N = 16$.

$$\begin{aligned} |x\rangle &\xrightarrow{\text{DFT}_{16}} \frac{1}{\sqrt{16}} \sum_{s=0}^{15} x_x(s)^* |s\rangle \\ \parallel \\ |x_0 x_1 x_2 x_3\rangle &= \frac{1}{\sqrt{16}} \sum_{s=0}^{15} \omega_{16}^{-sx} |s\rangle \end{aligned}$$

$$= \frac{1}{4} \left(|0000\rangle + \omega^{-x} |0001\rangle + \dots + \omega^{15x} |1111\rangle \right)$$

Unentangled form?

Amplitude on

$$|s_3 s_2 s_1 s_0\rangle : \frac{1}{4} \bar{\omega}^{s_2}$$
$$= \frac{1}{4} \bar{\omega}^{(s_0 + 2s_1 + 4s_2 + 8s_3)x}$$

If $s_0 = 1$ then phase shifts by $\bar{\omega}^x$

If $s_1 = 1$ then phase " " $\bar{\omega}^{2x}$

If $s_2 = 1$ " " " $\bar{\omega}^{4x}$

If $s_3 = 1$ " " " $\bar{\omega}^{8x}$

$$\left(\frac{|0\rangle + \bar{\omega}^{8x} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + \bar{\omega}^{4x} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + \bar{\omega}^{2x} |1\rangle}{\sqrt{2}} \right) \otimes$$

$$\frac{|0\rangle + \bar{\omega}^x |1\rangle}{\sqrt{2}}$$

$$= \frac{1}{4} \left(|0000\rangle + \bar{\omega}^x |0001\rangle + \bar{\omega}^{2x} |0010\rangle + \dots \dots \dots \right)$$

Challenge: Each output qubit depends on the whole of x .

Question: Does $|s_3\rangle = \frac{|0\rangle + \bar{\omega}^{8x} |1\rangle}{\sqrt{2}}$ depend on all of x_0, x_1, x_2, x_3 ?

$$\sum_{S_3=0}^1 \bar{\omega}^{8\pi S_3} \otimes$$

$$\sum_{S_2=0}^1 \bar{\omega}^{4\pi S_2} \otimes$$

$$\sum_{S_1=0}^1 \bar{\omega}^{2\pi S_1} \otimes$$

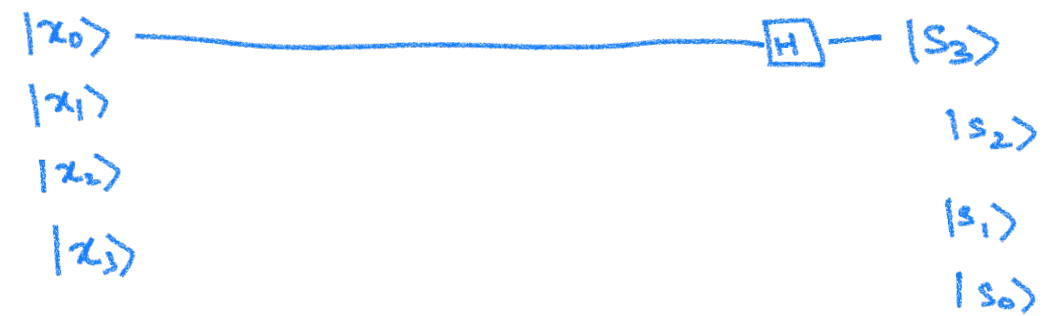
$$\sum_{S_0=0}^1 \bar{\omega}^{\pi S_0}$$

$$\frac{|0\rangle + \bar{\omega}^{8x} |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^{8(x_0+2x_1+4x_2+8x_3)} |1\rangle}{\sqrt{2}}$$

$$= \frac{|0\rangle + \bar{\omega}^{8x_0} |1\rangle}{\sqrt{2}}$$

$$= \frac{|0\rangle + e^{-\pi i x_0} |1\rangle}{\sqrt{2}}$$

$$\frac{|0\rangle + (-1)^{x_0} |1\rangle}{\sqrt{2}}$$



lets look at $|s_2\rangle$ now

$$\begin{aligned}
 |s_2\rangle &= \frac{|0\rangle + \bar{\omega}^4 |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^4 (x_0 + 2x_1 + 4x_2 + 8x_3)}{\sqrt{2}} \\
 &= \frac{|0\rangle + \bar{\omega}^{4x_0 + 8x_1} |1\rangle}{\sqrt{2}}
 \end{aligned}$$

$$\frac{|0\rangle + \bar{\omega}^{4x_0 + 8x_1} |1\rangle}{\sqrt{2}}$$

$$x_1 \text{ --- } [H] \text{ --- } \frac{|0\rangle + \bar{\omega}^{8x_1} |1\rangle}{\sqrt{2}} = |\phi\rangle$$



$$\begin{aligned}
 |00\rangle &\rightarrow |00\rangle \\
 |01\rangle &\rightarrow |01\rangle \\
 |10\rangle &\rightarrow |10\rangle \\
 |11\rangle &\rightarrow \bar{\omega}^4 |11\rangle
 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$|s_2\rangle = \frac{|0\rangle + \bar{\omega}^{4x_0} \bar{\omega}^{8x_1} |1\rangle}{\sqrt{2}}$$

$$|x_1\rangle \xrightarrow{H} \frac{|0\rangle + (-1)^{x_1} |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^{8x_1} |1\rangle}{\sqrt{2}}$$



$$|s_1\rangle = \frac{|0\rangle + \bar{\omega}^{2x} |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^{2x_0 + 4x_1 + 8x_2} |1\rangle}{\sqrt{2}}$$

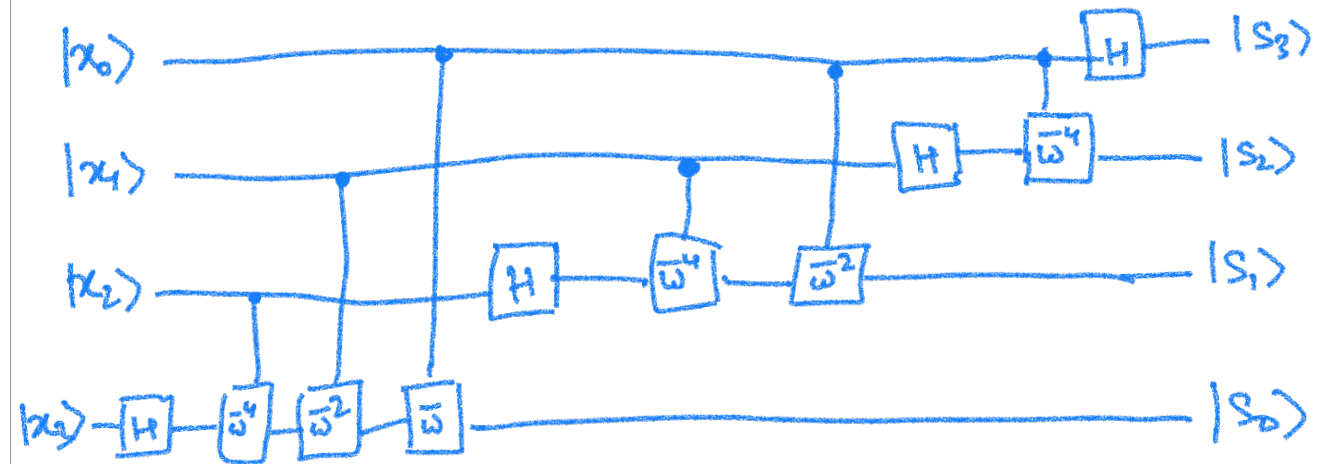
$$|x_2\rangle \xrightarrow{\text{H}} \frac{|0\rangle + (-1)^{x_2} |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^{8x_2} |1\rangle}{\sqrt{2}}$$

$$|s_1\rangle \leftarrow \frac{|0\rangle + \bar{\omega}^{-4x_1 + 8x_2} |1\rangle}{\sqrt{2}}$$

For getting $|S_1\rangle$ we need a gate

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{\omega^2} \end{bmatrix}$$

Final circuit



No. of gates $1+2+3+\dots+n = \frac{n(n+1)}{2} = O(n^2)$

Simon's algorithm for $\mathbb{Z}_N$.

$$N = 2^n$$

$$F : \mathbb{Z}_N \rightarrow \{0,1\}^m$$

is periodic with period L , i.e. $F(0), F(1), \dots, F(L-1)$

$$F(0) = F(L) = F(2L) = \dots$$

$$F(1) = F(L+1) = F(2L+1) = \dots$$

$$F(2) = F(L+2) = F(2L+2) = \dots$$

$\vdots$

FIND L (Assume L divides N)

are distinct

We will run the same algorithm with the correct Fourier transform.

$$|0^n\rangle \xrightarrow{\text{DFT}} \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle \xrightarrow{Q_F^*} \frac{1}{N} \sum_{x=0}^{N-1} |x\rangle \otimes |F(x)\rangle$$

$| \text{ancilla} \rangle$

The first step could be replaced by $H^{\otimes n}$. Why?

- Then we measure the last m qubits
- We will see $F(x)$ for a random x in $\{0, \dots, L-1\}$ with equal prob.
- State collapses to ?

$$\frac{1}{\sqrt{N/L}} (|x\rangle + |x+L\rangle + |x+2L\rangle + \dots)$$

We get for some random $x \in \{0, 1, \dots, L-1\}$

$$\frac{1}{\sqrt{N_L}} \left(|x\rangle + |x+L\rangle + \dots + |x + \left(\frac{N}{L} - 1\right)L\rangle \right) \otimes |F(x)\rangle$$

$\downarrow \text{DFT}_N$

$$\frac{1}{\sqrt{N_L}} \left(|x^*\rangle + |x^*_{x+L}\rangle + \dots + |x^*_{x + \left(\frac{N}{L} - 1\right)L}\rangle \right)$$

Amplitude on $|s\rangle$

$$\bar{\omega}^{sx} + \bar{\omega}^{s(x+L)} + \bar{\omega}^{s(x+2L)} + \dots$$

$$\bar{\omega}^{sx} \left(1 + \bar{\omega}^{sL} + \bar{\omega}^{2sL} + \dots + \bar{\omega}^{\left(\frac{N}{L}-1\right)sL} \right)$$

If sL is a multiple of N , then the
above sum is $\frac{N}{L} \cdot \bar{\omega}^{sx}$

If sL is not a multiple of N , then the sum is
0 (Check!)

So non-zero amp on

$$|0\rangle, \left|\frac{N}{L}\right\rangle, \left|\frac{2N}{L}\right\rangle, \dots$$

$$\text{Let } N/L = M$$

- On measurement, we get a random $|iM\rangle$
for a random $i \in \{0, 1, 2, \dots, L-1\}$

This gives iM for a random $i \in \{0, 1, \dots, L-1\}$
We repeat to get jM for a random $j \in \{0, 1, \dots, L-1\}$

CLAIM: $\gcd(iM, jM) = M$ w.p. $\geq \frac{1}{12}$

let p be a prime

$$P_p [p \text{ divides } i \text{ and } j] \leq \frac{1}{p^2}$$

$$P_2 [\text{g.c.d. is not } 1]$$

$$\leq P_2 [2 | \text{g.c.d.}] + P_2 [3 | \text{g.c.d.}] + \dots$$

$$\leq \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$\leq \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots$$

$$\leq \frac{1}{4} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \dots + \frac{1}{11^2} + \frac{1}{12^2} + \dots$$

$$\leq \frac{1}{4} + \frac{1}{3^2} + \frac{1}{3^2} + \frac{1}{3^2} + \frac{1}{6^2} + \frac{1}{6^2} + \dots + \frac{1}{6^2} + \frac{1}{12^2} + \dots + \frac{1}{12^2} + \dots$$

$$\leq \frac{1}{4} + \frac{3}{3^2} + \frac{6}{6^2} + \frac{12}{12^2} + \dots$$

$$= \frac{1}{4} + \frac{1}{3} \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) = \frac{1}{4} + \frac{2}{3} < 1$$

with const. prob we get M

$$L := \frac{N}{M}$$

Consider the case when L doesn't divide N

$$N \gg L$$

$$F(0) = F(L) = F(2L) = \dots = F\left(\left\lfloor \frac{N}{L} \right\rfloor \cdot L\right)$$

$$F(1) = F(L+1) = F(2L+1) = \dots$$

$$F(2) = F(L+2) = F(2L+2) = \dots$$

- Using the idea as Schmidt's we get for
a random $x \in \{0, 1, \dots, L-1\}$

$$\frac{1}{A} (|x\rangle + |x+L\rangle + |x+2L\rangle + \dots)$$

A is either $\lfloor \sqrt{N/L} \rfloor$ or $\lceil \sqrt{N/L} \rceil$

F.T. will give $\frac{1}{A} (|x^*\rangle + |x^*+L\rangle + \dots)$

Amplitude on $|s\rangle$

$$\bar{\omega}^s x + \bar{\omega}^{s(x+L)} + \bar{\omega}^{s(x+2L)} + \dots$$

CLAIM: This amplitude is large when

s is $\lfloor \frac{N}{L} \rfloor$ or $\lfloor \frac{2N}{L} \rfloor$ or $\dots$

Let s be $\lfloor iN/L \rfloor$ i.e. integer in $(\frac{iN}{L} - \frac{1}{2}, \frac{iN}{L} + \frac{1}{2}]$

The amp. is

$$\bar{\omega}^{sx} \left[1 + \bar{\omega}^{Ls} + \bar{\omega}^{2Ls} + \dots \right]$$

$$\bar{\omega}^{sx} \left[1 + \bar{\omega}^{L \left(\frac{iN}{L} \pm \frac{1}{2} \right)} + \bar{\omega}^{2L \left(\frac{iN}{L} \pm \frac{1}{2} \right)} + \dots + \bar{\omega}^{kL \left(\frac{iN}{L} \pm \frac{1}{2} \right)} \right]$$

$$k = \lfloor N/L \rfloor - 1$$

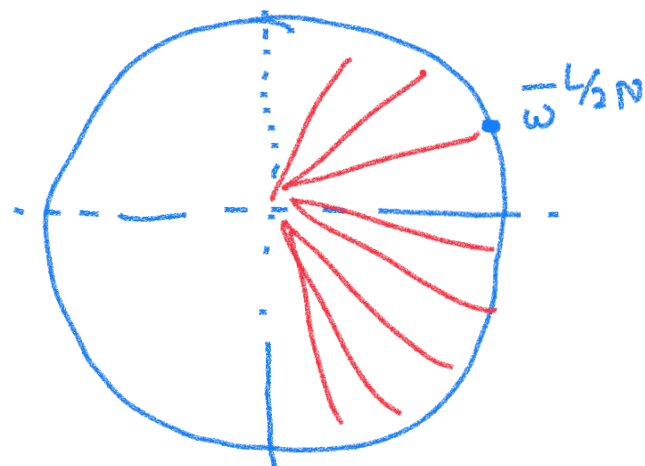
When we say $\pm \frac{1}{2}$, this is shorthand for
a no. between $(-\frac{1}{2}, \frac{1}{2}]$

$\pm \frac{L}{2}$ is shorthand for a number
between $(-\frac{L}{2}, \frac{L}{2}]$

$$\bar{\omega}^{sx} \left[1 + \bar{\omega}^{\pm L/2} + \bar{\omega}^{\pm \frac{2L}{2}} + \dots + \bar{\omega}^{\pm \left\lfloor \frac{N}{L} - 1 \right\rfloor \frac{L}{2}} \right]$$

$$\bar{\omega} = e^{2\pi i/N}$$

The x -coordinate
of the amplitude
does not cancel



We will not see a proof of the claim
but the amplitudes will "cancel much more"
when we are farther from $\lfloor iN/L \rfloor$ for
any i .